

Collections

Introduction to Programming

StayAhead Training; Sep, 2024

Introduction

Collections

- **Collection:**
 - A group of things in memory
 - Each member is accessible via the group
- A plural noun is indicative of a collection, e.g. clients, results, etc.

Arrays

Collections

- **Array:** the simplest and most common type of collection
- Arrays are indexed; each element is accessed via its index
- In some languages the size of an array is fixed

Array Creation

Collections

- Pseudocode, e.g.:
 - SET Names to ["Tom", "Dick", "Harry"]

Array Creation

Collections

- PY: `numbers = [2, 4, 6, 8]`
- _____
- JS: `var numbers = [2, 4, 6, 8];`
- _____
- JV: `int[] numbers = {2, 4, 6, 8};`
- _____
- C#: `int[] numbers = {2, 4, 6, 8};`

Array Creation

Collections

- PY: `names = ["James", "George", "Olivia"]`
- -----
- JS: `var names = ["James", "George", "Olivia"];`
- -----
- JV: `String[] names = {"James", "George", "Olivia"};`
- -----
- C#: `string[] names = {"James", "George", "Olivia"};`

Arrays with Various Types of Data Collections

- PY: `things = ["James", 123, True]`
- -----
- JS: `var things = ["James", 123, true];`
- -----
- JV: `Object[] things = {"James", 123, true};`
- -----
- C#: `object[] things = {"James", 123, true};`

Getting Elements from an Array

Collections

- Pseudocode, e.g.:
 - SET FirstName to Names[0]
 - OUTPUT FirstName to stdout
- Or, simply:
 - OUTPUT Names[0] to stdout

Getting Elements from an Array

Collections

- PY: `third_number = numbers[2]`
- _____
- JS: `var thirdNumber = numbers[2];`
- _____
- JV: `int thirdNumber = numbers[2];`
- _____
- C#: `int thirdNumber = numbers[2];`

Setting Elements in an Array (initial state)

Collections

Index	Element
0	2
1	4
2	6
3	8

Setting Elements in an Array

Collections

- Pseudocode, e.g.:
 - SET Names[0] to "Thomas"

Setting Elements in an Array

Collections

- PY: `numbers[2] = 7`
- `_____`
- JS: `numbers[2] = 7;`
- `_____`
- JV: `numbers[2] = 7;`
- `_____`
- C#: `numbers[2] = 7;`

Setting Elements in an Array (after setting)

Collections

Index	Element
0	2
1	4
2	7
3	8